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Resonant ultrasound spectroscopy measurement and modeling of additively manufactured octet truss lattice cubes

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1. Introduction

Additive manufacturing (AM) can produce complex parts with internal features that would be impossible to produce by conventional manufacturing. Metal lattices are a quintessential application of AM to achieve lower density than bulk metal parts and advanced design methods such as strut topology optimization. In particular, the octet truss lattice geometry is desirable for its high strength and stiffness. To realize the competitive advantage of AM lattices requires a high degree of quality control. Defects such as missing and disconnected struts that form during the AM process [1,2] are a hindrance to the reproducibility and performance of lattices [1,3,4]. Characterization of lattices can be challenging due to the significant size difference between lattice features such as struts and the overall part size. Two non-destructive techniques for characterization of lattices are X-ray computed tomography (CT) and resonant ultrasound spectroscopy (RUS).

X-ray CT is a non-destructive three-dimensional (3D) imaging technique that is capable of interrogating both external and internal features to a high resolution. Drawbacks of X-ray CT are the time and specialized equipment required for scanning, image reconstruction, and analysis. RUS is an acoustic technique that is sensitive to defects, geometric changes, and material phase [5–7]. It tracks fundamental mechanical resonance modes and provides rapid holistic probing of a volume [8]. RUS provides a high throughput technique that is complementary to X-ray CT, whereby RUS could be utilized to screen AM parts for abnormalities that would be further evaluated by X-ray CT. Recent literature has applied RUS to parts manufactured by the AM technique laser powder bed fusion such as nickel alloy dogbones [5], aluminum alloy cylinders [6], and stainless steel rhombic dodecahedron lattice cubes [8].

In the present work, RUS and X-ray CT were used to characterize additively manufactured Ti-5553 lattice cubes containing various percentages of built-in missing struts. Compression tests were conducted to investigate the elastic modulus with respect to percentage of missing struts. Finite element models of the lattice cubes were developed using measured properties from the additively manufactured parts. Eigenfrequencies from the model were compared to experimental RUS measurements of resonance frequencies.

2. Material and methods

2.1 Additively manufactured specimens

Octet truss lattice specimens were built using Ti-5553 powder (AP&C lot 201-D0030) on a SLM 280 laser powder bed fusion machine. Nominal designs of overall specimen geometry and unit cell for the octet truss lattice cubes are shown in Figure 1. Each specimen was composed of a 9x9x9 lattice of octet truss unit cells, and two solid skins of 5 mm thickness on two opposing faces. Nominal strut diameter and unit cell edge length were respectively 424 µm and 4.56 mm. The square face of each skin encompasses the lattice along the x and y axes. Each skin overlaps the lattice face by half a strut diameter along the z axis to ensure fusion during printing,

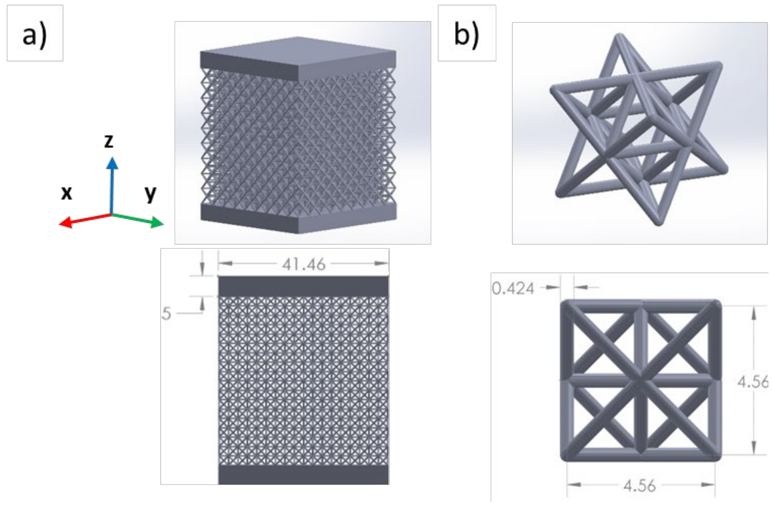

Figure 1. Isometric and front views of nominal designs for the (a) overall specimen geometry and (b) octet truss lattice unit cell. Dimension units are millimeters.

All specimens were produced on one build plate with build direction along the y axis depicted by the Cartesian coordinate system in Figure 1(a). To prevent the lattice from being in direct contact with the build plate along the x-z plane, a sacrificial skin of 2 mm thickness was built between the specimens' x-z plane and the build plate. The as-built specimens were cleaned by ultrasonic rinsing in deionized water and stress relieved on the build plate at 572°F for a soak time of one hour and force cooled using argon gas. Each specimen was separated from its sacrificial skin and removed from the build plate using electrical discharge machining. The mass of each specimen was measured using an analytical balance (Mettler Toledo XP204). The length, width, height, and skin thicknesses of each specimen were measured using a dial caliper (Mitutoyo D6TX). Table 1 summarizes the mass and dimensions of the specimens.

Table 1. Measurements from overall specimens and isolated octet truss lattices. ρ_c is the effective density of each isolated octet truss lattice's bounding box.

Specimen name	Overall specimen mass (g)	Skin thickness (mm)		Lattice bounding box dimensions (mm)			Lattice mass (g)	ρ_c (kg/m^3)
		Top	Bottom	x axis	y axis	z axis		
0% (#1)	94.7381	5.019	5.007	41.316	41.266	40.944	16.9761	243.186
0% (#2)	94.9528	5.019	5.008	41.334	41.227	41.050	17.2002	245.888
0% (#3)	95.0685	4.970	4.987	41.342	41.229	41.048		
0.5% (#1)	94.6323	4.987	4.989	41.365	41.239	41.024	16.7451	239.335
0.5% (#2)	96.7061	4.985	4.997	41.348	41.173	41.059	18.9581	270.907
0.5% (#3)	94.3233	4.989	5.004	41.318	41.257	41.013		
1% (#1)	93.9170	4.981	4.995	41.288	41.135	41.055	16.4709	235.637
1% (#2)	93.9047	5.004	5.008	41.325	41.226	40.992	16.3632	234.056
1% (#3)	94.1933	5.004	4.997	41.342	41.198	40.982		

Missing struts were designed into some of the specimens. Three nominal missing strut percentages were used for this study: 0%, 0.5%, and 1%. A computer-aided design (CAD) file was first created for the 0% missing strut specimen geometry. For the 0.5%, and 1% missing strut geometries, the 0% missing strut CAD file was modified to remove struts from the 9x9x9 octet lattice. The missing strut locations were randomly generated. Three specimens were then built for each of the designs. For example, specimens 0.5% (#1), 0.5% (#2), and 0.5% (#3) were all built using the same CAD file.

2.2 X-ray computed tomography

The 9x9x9 octet truss lattice of each specimen was inspected using a North Star Imaging X3000 industrial CT X-ray inspection system. The system was run at 190 kV and 450 µA, equivalent to 85.5 W of power, with a 0.08” copper filter. A 0.2 beam hardening coefficient and ring reduction feature were used in accordance with a proprietary algorithm from the manufacturer. A helical scan path, defined by a height of 180 mm and four stage revolutions per scan, enabled inspection of the lattice portion of each specimen with minimal shadowing effect from the solid skins. Magnification was 2.19X and acquisition rate was 10 frames per second. Reconstruction data was exported as two-dimensional (2D) image slices of the specimen volume along the AM build direction. No digital filter was used during reconstruction, and cubic voxel size was 58.1 µm.

Analysis of the reconstruction data was conducted using a MATLAB routine. The initialization of the routine defines a model octet truss coordinate system using inputs for strut length, strut diameter, and voxel size. The reconstructed image slices for each specimen are imported to MATLAB and transformed to align as closely as possible to the model coordinate system. The transformations consist of geometrical adjustments such as shifts, rotations, and scaling. The image slices are also segmented to remove background noise using a histogram shape-based thresholding method.

The routine then extracts a sub-volume of the reconstruction data for each expected strut location in the model coordinate system. Properties of each sub-volume, such as bounding box and centroid, are calculated. 2D slices are generated for various planes of the 3D transformation, along with two orthogonal projections of each sub-volume. Figure 2 displays an example of projections for one 2D slice of specimen 0.5% (#1). Identified disconnected and missing struts are examined further by orthogonal projections of Figure 2(b) and (c). The top projection indicates a view parallel to the 2D slice, i.e., Figure 2(a), whereas the side projection indicates a view orthogonal to the 2D slice. Sub-volumes are flagged as disconnected or missing struts based on criteria to check continuity, such as gaps in pixel intensity and deviation from the model octet truss coordinate system. All the flagged sub-volumes were manually reviewed to confirm their classification as either disconnected or missing struts.

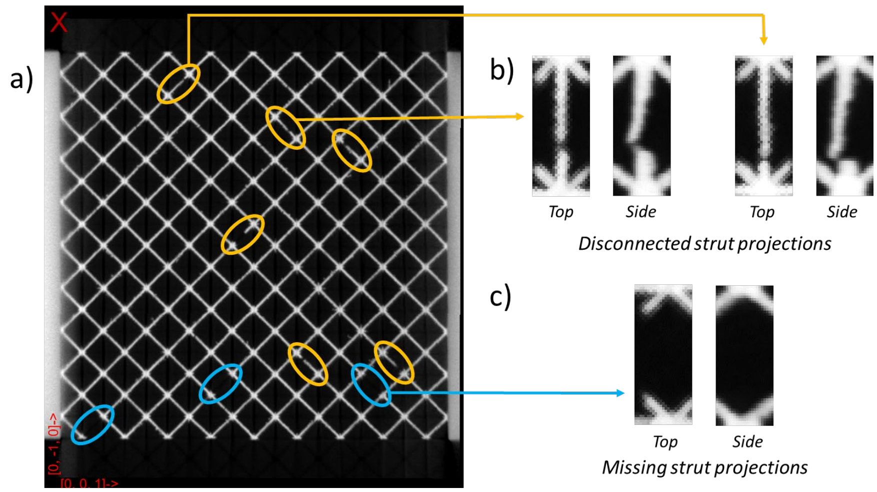

Figure 2. (a) Example of 2D slice from X-ray CT reconstruction data analysis of specimen 0.5% (#1) with identified (b) disconnected struts and (c) missing struts. Top and side projections of example strut sub-volumes are shown.

2.3 Resonance frequency experimental measurement

Instrumentation for ultrasound spectrum measurement comprised of two piezoelectric transducers (Dunegan Research Corporation S-140A), a waveform generator (Keysight 33500 B Series), and a lock-in amplifier (Stanford Research Systems SR865A). Figure 3(a) schematically illustrates the measurement system. The waveform generator produces a drive signal of constant frequency to the top transducer. The same drive signal is also sent to the lock-in amplifier. The specimen is held between the two transducers by a pair of its opposing vertices to minimize coupling losses [9]. Transmission of the drive signal results in an elastic response from the specimen that is received by the bottom transducer and sent to the lock-in amplifier as an output signal. The drive signal and output signal are compared in the lock-in amplifier, where the product of in-phase and quadrature relationships is calculated as an amplitude of arbitrary units. This entire process is repeated for a range of frequencies from the waveform generator, with the result being an amplitude spectrum. Figure 3(b) shows an example spectrum measurement for the 0% (#1) specimen. In the spectrum, a local maxima of high Q factor represents a resonance mode of the specimen.

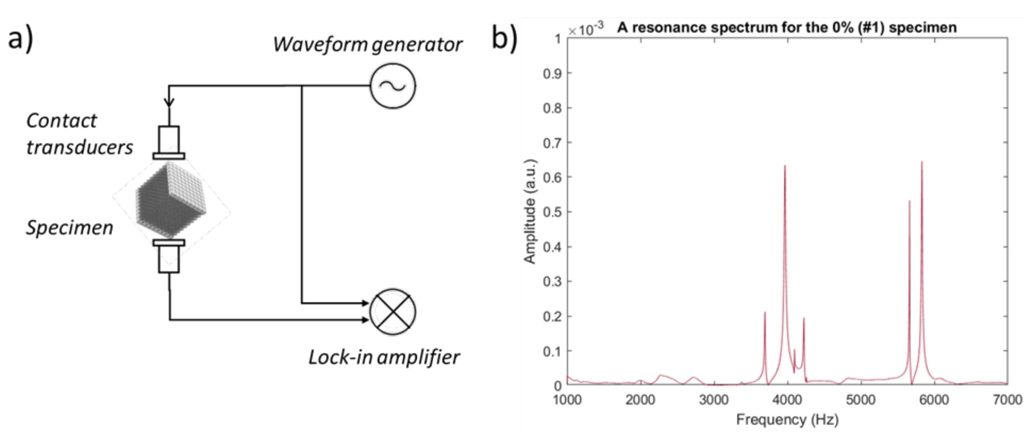

Figure 3. (a) Schematic of resonant ultrasound spectroscopy system and (b) example of an experimental resonance spectrum measurement.

Settings for the waveform generator were chosen to sufficiently capture the resonance response from the specimen geometry and optimize the output signal quality. The generated signal was a standard sine wave of peak-to-peak voltage 10 V, swept over a frequency range from 1 kHz to 7 kHz using a step size of 2 Hz. Each specimen had its resonance response measured five times to ensure reproducibility of the resonance frequencies. In between each measurement, the specimen was repositioned by selecting another pair of its opposing vertices to be contacted by the transducers. All measurements were conducted at room temperature.

2.4 Finite element modeling

Two finite element models of the overall specimen geometry were created using COMSOL Multiphysics 5.6. Figure 4(a) shows the nominal model, which replicates the nominal design of the overall specimen geometry. The nominal model was created by assembling a 9x9x9 array of unit cells (Figure 1(b)). The array along with the top and bottom skins were united into a single object using the Form Union operation. Figure 4(b) shows the continuum model, which replaces the octet truss lattice with a solid continuum approximation of its bounding box.

Free boundary conditions were imposed on all the outer surfaces in both models. A tetrahedral mesh and the multifrontal massively parallel sparse (MUMPS) direct linear system solver provided in COMSOL were used to calculate resonance frequencies and mode shapes through the software's Eigenfrequency solver. The mesh size was set to Normal for the nominal model.

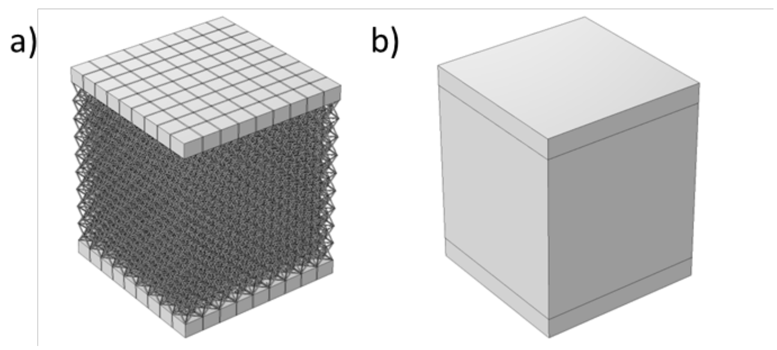

Figure 4. Finite element model geometries. (a) Nominal model that replicates the overall specimen geometry nominal design. The square grid on the skin indicates the Array operation used in recreating the nominal geometry. (b) Continuum model that substitutes a solid continuum approximation of the octet truss lattice.

Table 2 summarizes the material properties applied in the finite element models. Material properties of room temperature bulk Ti-5553 were applied to the entire nominal model and to the skins of the continuum model. Effective properties of the octet truss lattice bounding box were applied to the solid continuum. These effective properties comprised of density, ρ_c, Poisson's ratio, ν_c, Young's modulus, E_c, and elasticity matrix, D_c.

Table 2: Summary of material property inputs to the finite element models.

	Density (kg/m^3)	Poisson's ratio	Young's modulus (GPa)	Elasticity matrix
Bulk Ti-5553	4650 [10,11]	0.310 [12]	85 [12]	Isotropic
Lattice bounding box	ρ_c [Table 1]	ν_c	E_c	D_c

To measure ρ_c, electrical discharge machining was used to remove the skins and isolate the octet truss lattices from six specimens: 0% (#2), 0% (#3), 0.5% (#2), 0.5% (#3), 1% (#2), and 1% (#3). Mass of the isolated octet truss lattices was measured using an analytical balance (Mettler Toledo XP204). Length, width, and height of the isolated octet truss lattices were measured using a dial caliper (Mitutoyo D6TX). Effective density of the octet truss lattices was calculated as the quotient of mass and bounding box volume as approximated by the product of the length, width, and height measurements.

Relationships for ν_c, E_c, and D_c with respect to elastic constants were adopted from analytical solutions of the octet truss geometry [13,14]. Equations 1 – 3 identify how these relationships were defined in the continuum model.

$$D_c = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{12} \end{bmatrix} \tag{1}$$

$$C_{12} = E_c \left(\frac{(1-\nu_c)}{\nu_c} + 1\right) \left(\frac{(1-\nu_c)}{\nu_c} - 2\right)^{-1} \tag{2}$$

$$C_{11} = C_{12} \frac{(1-\nu_c)}{\nu_c} \tag{3}$$

The effective properties ν_c and E_c were optimized in the continuum model to best fit experimentally measured resonance frequencies. A Monte Carlo solver was used to parametrically sweep E_c and ν_c respectively between the bounds of 0.01 – 2 GPa and 0 – 1. The bounds for E_c were selected to encompass values expected from analytical relationships of octet truss lattices [15]. Equation 4 defines the objective function, δ, used for minimization.

$$\delta = \sum_i \left| f_i^{fit}(E_c, \nu_c) - f_i^{meas} \right| \tag{4}$$

The index, i, spans the number of resonance frequency modes to be fitted where f_i^{meas} is the ith experimentally measured resonance frequency. The resonance frequency values for f_i^{fit} were calculated by the Eigenfrequency solver for iterative combinations of E_c and ν_c until convergence is achieved according to the default optimality tolerance of 0.01. The continuum model mesh size setting was Extremely coarse to minimize computational time for the Monte Carlo solver, which resulted in a rough estimate of the best-fit combination of E_c and ν_c. For further refinement, the continuum model mesh size was set to Finer and a Nelder-Mead solver was used with initial guesses being the best-fit values of ν_c and E_c determined from the Monte Carlo solver.

2.5 Uniaxial compression testing

Three specimens, 0% (#3), 0.5% (#3), and 1% (#3), were compression tested on an Instron 8562 system. These specimens were preserved in the geometry of Figure 1(a), and thus did not have their skins removed for isolated lattice mass and effective density measurements. The specimen was placed on the bottom platen and the top platen was lowered onto the top surface of the specimen until a preload of 1 N was measured. The crosshead displacement rate was 0.015 in/min as measured by the mean of three MTS 632.128-20 extensometers. All tests were conducted at room temperature.

All three compression tests consisted of loading to a maximum force of 1400 N followed by unloading. Specimen 1% (#3) was unloaded only once to 0 N. To confirm the repeatability of the modulus measurement, specimens 0% (#3) and 0.5% (#3) were unloaded to 200 N followed by reloading to 1400 N and unloading to 0 N. Effective engineering stress and engineering strain acting on each entire specimen were calculated using the measurements of load, crosshead displacement, and bounding box cross-sectional area in the x-y plane of the specimen. The bounding box cross-sectional area was assumed to remain constant throughout the test. The Young's modulus of each entire specimen was calculated using linear fits of the engineering stress versus engineering strain plot between 500 kPa and peak stress.

3. Results and Discussion

3.1 Specimen inspection

3.1.1 Quantification of missing and disconnected struts

Table 3 displays results from X-ray computed tomography analysis of missing and disconnected struts. The classifications of missing struts and disconnected struts are mutually exclusive such that missing struts do not count towards the disconnected struts. The measured disconnected strut percentage is consistently 5% ± 0.25%, with two slight exceptions being 3.89% for specimen 0% (#1) and 5.72% for specimen 1% (#3). The measured missing strut percentage for all specimens is higher than the nominal missing strut percentage by 0.02% - 0.12%. This trend indicates that missing struts were unintentionally formed during the manufacturing process [1] or from handling of the parts [2,3].

Table 3. Measurements of missing and disconnected strut percentages from X-ray computed tomography analysis.

Specimen name	Missing strut %		Disconnected strut %
	Nominal	CT	
0% (#1)	0%	0.04%	3.89%
0% (#2)		0.02%	4.75%
0% (#3)		0.04%	5.26%
0.5% (#1)	0.50%	0.57%	4.97%
0.5% (#2)		0.62%	5.09%
0.5% (#3)		0.56%	5.13%
1% (#1)	1%	1.07%	5.04%
1% (#2)		1.11%	4.82%
1% (#3)		1.05%	5.72%

3.1.2 Mass and dimensions

The measurements summarized in Table 1 indicate that the skin thicknesses and lattice bounding box dimensions amongst the specimens are consistent. Mass of the overall specimens and isolated lattices trends lower with increasing percentage of missing struts. The one outlier to this trend is specimen 0.5% (#2), which is at least 1.75 g more massive than any other specimen. The difference in masses is consistent between the overall specimen and isolated lattice measurements, which indicates that the mass variation occurs in the lattice of each specimen and not the solid skins. Mass variation can be indicative of a discrepancy dependent on the build plate position of each specimen, for example due to gas flow directionality [16], recoater blade non-uniformity, or other anomalies [17].

3.2 Resonance frequency measurements

Table 4 displays mean values of experimental resonance frequency measurements from all the overall specimens. Two doublets are observed amongst the resonance frequency modes, the first being modes 2 and 3, and the second being modes 5 and 6. Doublets are susceptible to overlapping in experimental measurements, which can result in one of the doublet modes being undetected [18]. Mode 3 was not detected in five of the nine specimens, whereas modes 2, 5, and 6 were detected in all the specimens.

The mean resonance frequencies in Table 4 trend lower for the 0.5% and 1% missing strut specimens compared to the 0% missing strut specimens. Missing struts reduce the effective density and stiffness of truss lattices [4,19], with both effects shifting resonance modes to lower frequencies [5–7]. The one outlier to these trends is specimen 0.5% (#2), which has resonance frequency values closer to the 1% missing strut specimens than the 0.5% specimens. The resonance frequency for mode 5 is especially anomalous for specimen 0.5% (#2), as it is lower than all the 1% specimens' mode 5 frequencies by at least 90 Hz. Recall from Table 1 that specimen 0.5% (#2) is also an outlier by having a lattice mass significantly larger than all other specimens. Larger mass, while resulting in higher effective density of the lattice bounding box, may not result in a higher stiffness. For specimen 0.5% (#2), its lower resonance frequencies indicate that its larger mass has coincided with a lower effective stiffness of the lattice. This reduction in stiffness and resonant frequency may be caused by a deviation to the lattice microstructure [5], for example resulting from slower cooling rates in thicker struts.

Table 4. Mean values of experimentally measured resonance frequencies.

Specimen name	Mean resonance frequency (Hz)					
	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
0% (#1)	3697.8	3991.3	4138.5	4233.7	5659.2	5849.7
0% (#2)	3775.9	3999.5	*Undetected*	4281.5	5664.3	5866.3
0% (#3)	3849.6	4045.1	*Undetected*	4357.7	5771.6	6005.7
0.5% (#1)	3651.6	3945.1	*Undetected*	4144.0	5550.7	5771.9
0.5% (#2)	3563.0	3830.5	3967.2	4026.3	5238.4	5673.1
0.5% (#3)	3723.9	3977.3	*Undetected*	4182.0	5568.6	5732.4
1% (#1)	3737.0	3969.3	*Undetected*	4176.6	5535.1	5703.4
1% (#2)	3694.7	3783.9	3896.8	3992.0	5332.1	5493.4
1% (#3)	3597.3	3764.8	3841.8	4148.0	5407.1	5630.6

3.3 Best-fit resonance mode shapes and frequencies

Optimization of ν_c and E_c in the continuum model was performed on the specimens 0% (#1), 0.5% (#2), and 1% (#2). These three specimens were chosen because they all possessed a complete set of detected resonance frequency modes 1 – 6 and ρ_c measurements.

Table 5 displays the best-fit values for ν_c and E_c resulting from the continuum model optimization. The best-fit values of ν_c are consistent amongst the three specimens and are similar to values found in a computational study on octet truss lattices [15]. The best-fit values of E_c trend lower with increasing missing strut percentage, which agrees with the trend inferred from the experimental resonance frequency measurements. The E_c of specimen 0.5% (#2) is lower than that of specimen 0% (#1), confirming that the outlier large ρ_c of specimen 0.5% (#2) coincided with a lower effective lattice stiffness.

Table 5. Effective Poisson's ratios, ν_c, and effective Young's moduli, E_c, that best-fit the continuum modeled resonance frequencies to experimentally measured resonance frequencies.

Specimen name	ν_c	E_c (GPa)
0% (#1)	0.342	0.342
0.5% (#2)	0.344	0.321
1% (#2)	0.339	0.306

Figure 5 displays modeled resonance mode shapes with overlay visualizations of surface displacement where antinodes are darker regions and nodes are indicated by lighter regions. The mode shapes from the nominal model and the continuum model are respectively Figure 5(a) and Figure 5(b). The order of resonance mode shapes resembles those of a rhombic dodecahedron lattice cube part [8], with mode 1 being torsional, modes 2 and 3 being flexural, mode 4 being breathing, and modes 5 and 6 being shear. The order and identification of mode shapes are consistent between the nominal model and the continuum model. The octet truss lattice has a minor effect on the resonance mode shape, as observed by antinodes being on the specimen corners and edges in the nominal model versus along the specimen centerline planes in the continuum model.

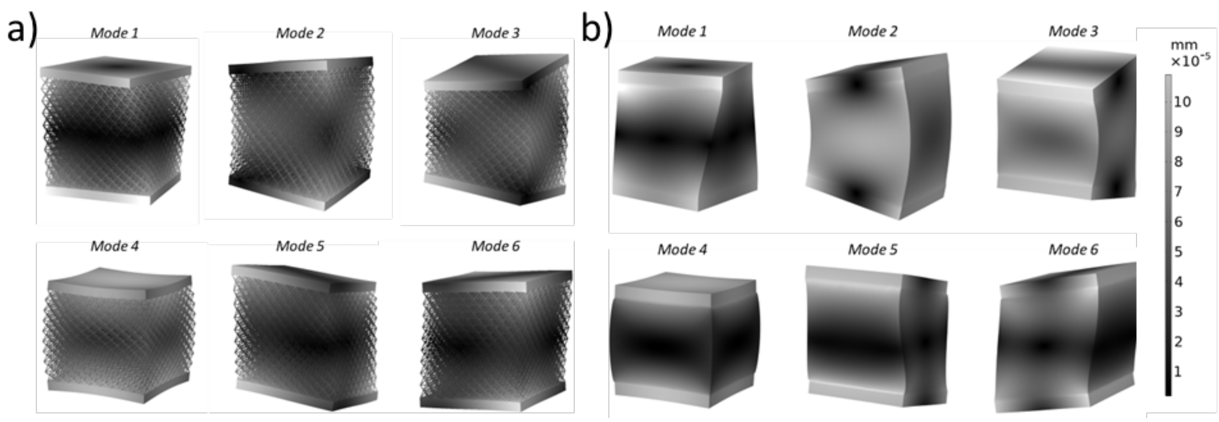

Figure 5. Surface displacement visualization of the resonance modes from (a) the nominal model and (b) the continuum model using best-fit ν_c and E_c

Table 6 displays resonance frequencies from the nominal model and best-fit resonance frequencies from the continuum model, with δ_i being the difference between experimental and model for each resonance frequency. Near-zero values of δ_i indicate excellent accuracy for modes 2 and 6 of the continuum model. The other modes have reasonable δ_i with the largest magnitude being the -490.7 Hz for specimen 1% (#2) mode 1. The magnitude and sign of δ_i for all modes is consistent amongst all the modeled resonance frequencies. This trend indicates the effect of a systematic difference between the experimentally measured specimens and the modeled geometries, for example the disconnected struts or unintentional missing struts. The overall consistency between the mode shapes in Figure 5 and the modeled resonance frequencies in Table 6 confirms the feasibility of the continuum model as a simplified representation of the nominal octet truss lattice geometry.

Table 6. Resonance frequencies calculated from the nominal model and from the continuum model using best-fit ν_c and E_c. The difference between each best-fit resonance frequency and its corresponding experimentally measured resonance frequency is shown in the rows labelled δ_i.

Specimen name	Resonance frequency (Hz)					
	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
Nominal model	3531.4	4118.6	4118.7	4316.9	5693.3	5693.4
δ_i	-166.4	+127.3	-19.8	+83.2	+34.1	-156.3
0% (#1)	3426.3	3991.6	3992.7	4284.7	5843.3	5849.1
δ_i	-271.5	+0.3	-145.8	+51.0	+184.1	-0.6
0.5% (#2)	3335.7	3830.8	3831.9	4133.4	5666.7	5672.3
δ_i	-227.3	+0.3	-135.3	+107.1	+428.3	-0.8
1% (#2)	3204.0	3784.3	3785.3	4057.7	5486.0	5491.5
δ_i	-490.7	+0.4	-111.5	+65.7	+153.9	-1.9

3.4 Compression response characteristics

Figure 6 plots the stress–strain response measured from compression tests on the three specimens with the preserved overall geometry of Figure 1(a). Hysteresis is observed for all three results with non-linear behavior during initial loading followed by linear behavior throughout unloading and reloading cycles. The non-linear behavior may be due to localized yielding of the lattice [20] or bedding-in effects at the lattice nodes [15].

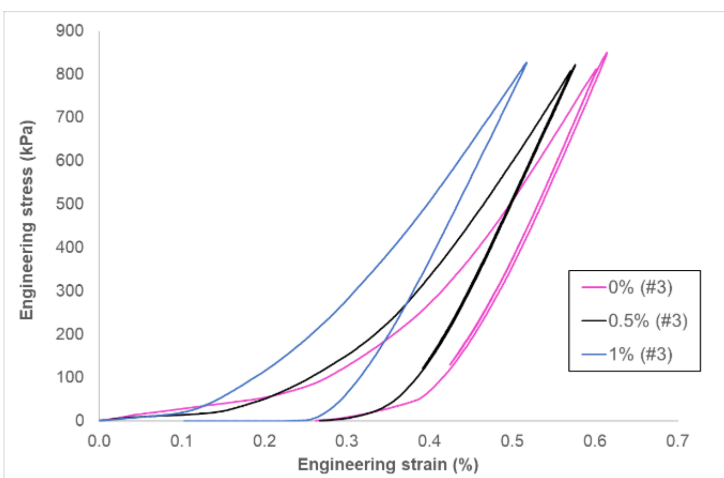

Figure 6. Plots of engineering stress and engineering strain data from compression tests.

Table 7 displays values of Young's modulus calculated using linear best-fits to the stress–strain data from 500 kPa to the peak stress of each test. The 500 kPa lower bound was chosen as an estimate of the transition from non-linear to linear behavior during initial loading. The moduli for both initial loading and unloading decrease with increasing percentage of missing struts, which confirms the trend observed from the best-fit finite element models. The unloading modulus is larger than the initial loading modulus by approximately 0.12 GPa, but together they serve as nearby bounds for the E_c values from the continuum model. The unloading modulus being an upper bound is indicative of contribution from the solid skins to the elastic response during the compression test. The initial loading modulus being a lower bound indicates localized yielding or bedding-in of the lattice that is not observed upon unloading and reloading and is not captured by the finite element model which assumes linear elasticity.

Table 7. Young's moduli as calculated from linear fits to the compression test data from 500 kPa to peak stress.

Specimen name	Young's modulus (GPa)	
	Initial loading	Unloading
0% (#3)	0.301	0.439
0.5% (#3)	0.292	0.413
1% (#3)	0.275	0.390

4. Conclusion

Increasing missing strut percentage in the specimens was correlated to trends of decreasing lattice mass, decreasing stiffness, and decreasing resonance frequencies of the overall specimen geometry. The effective Young's modulus of the lattice bounding box was determined using optimized finite element models of the specimen geometry, and reasonable accuracy was confirmed by compression testing. A solid continuum approximation of the octet truss lattice produced agreement with the nominal geometry in terms of modeled mode shapes and resonance frequencies. A systematic difference was observed between the resonance frequencies of the experimentally measured specimens and the modeled geometries. The difference can be traced to disconnected struts and unintentional missing struts in the specimens as quantified by X-ray CT analysis. One specimen possessed an outlier lattice that was at least 1.75 g more massive than any other specimen. Despite its larger ρ_c, the outlier lattice mass coincided with a lower effective Young's modulus as detected by experimental resonance frequency measurements and finite element model parameter optimization.

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